

connecting things

how to (in hardware) connect A and B?

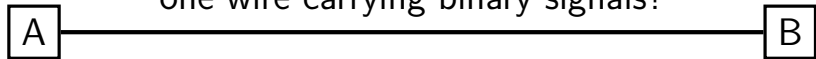
A

B

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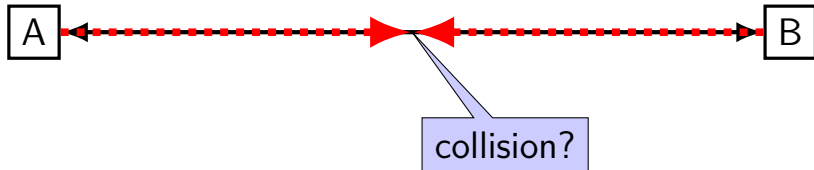
how to (in hardware) connect A and B?

one wire carrying binary signals?



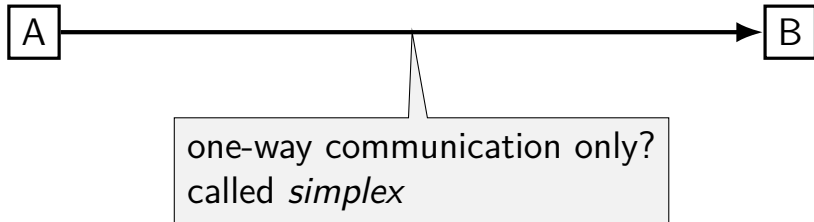
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how to (in hardware) connect A and B?



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...and later



taking turns, but one-way
called *half-duplex*
challenge: how to agree who's turn?

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how to (in hardware) connect A and B?



both ways at the same time
called *full duplex* (or *duplex*)

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how to (in hardware) connect A and B?



here: duplex via multiple wires (simplest scheme)
can achieve effect electrically/etc. via one wire
example: cable Internet
(how is topic for ECE class)

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A

B

C

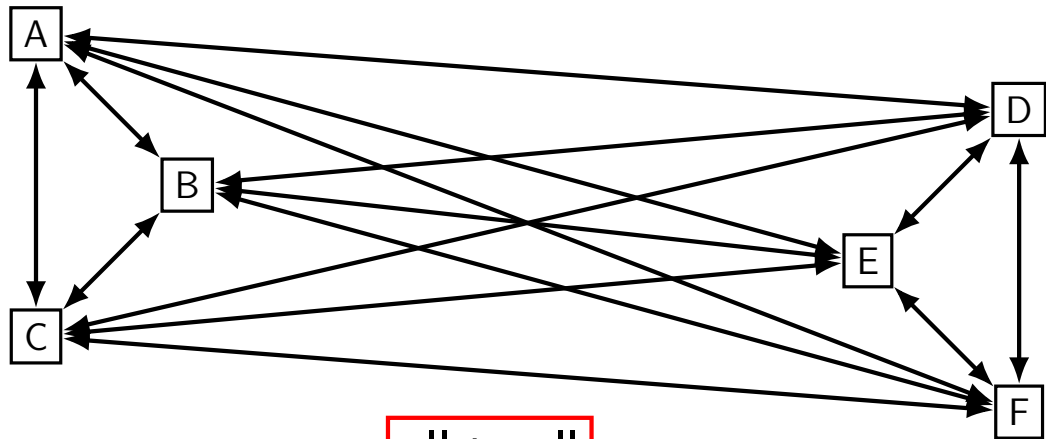
D

E

F

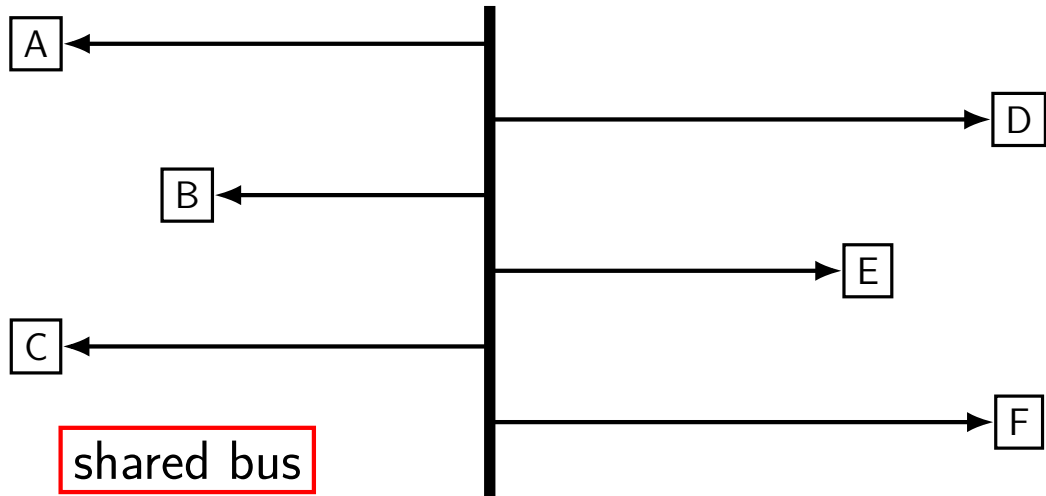
how to connect?

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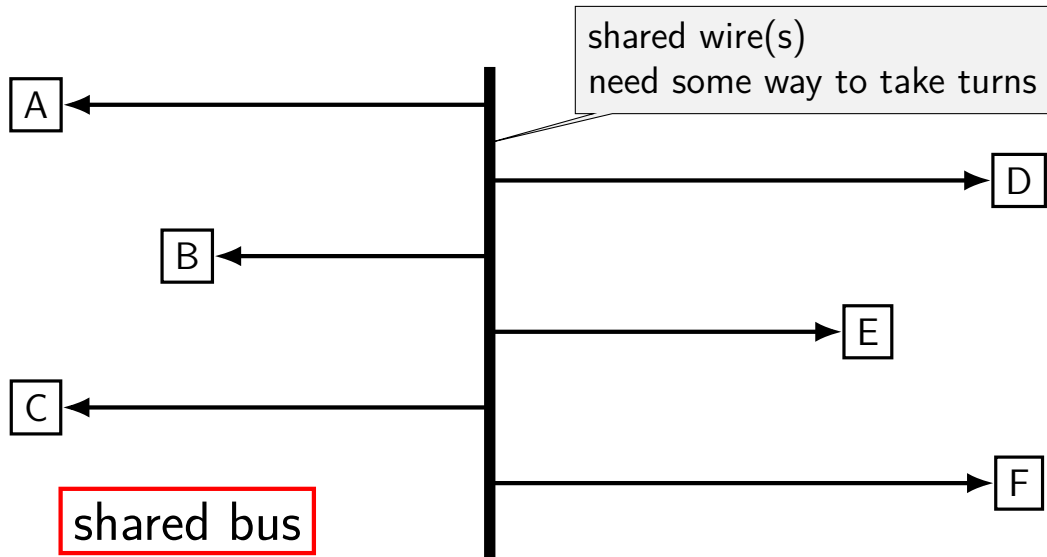


all-to-all

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shared bus, really?

common for parts of internals of computers (topic later)

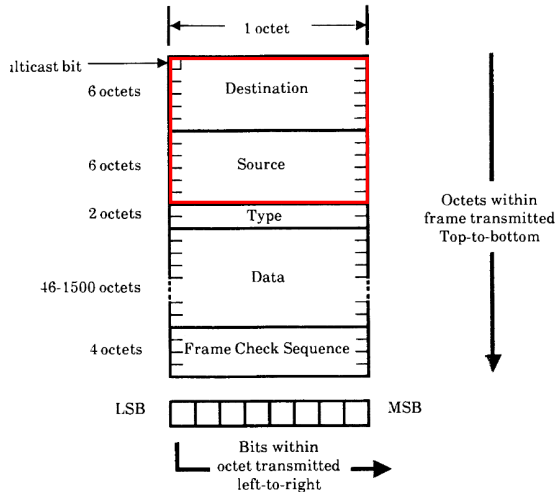
model for wifi

radio “channel” kinda similar to shared wire

how the early versions of Ethernet worked

“vampire taps” physically attached to shared cable

shared bus, messages for who?



messages need a '*header*' to tell who it's to/from

everyone needs to filter out messages that aren't theirs

Figure 6-1: Data Link Layer Frame Format

taking turns on shared bus?

token ring

- one machine has a 'token' = can send
- send special message to pass to another machine

free-for-all: collision detection + retry

- detect if you're transmitting when someone else is
- wait (usually randomized amount of time) and retry

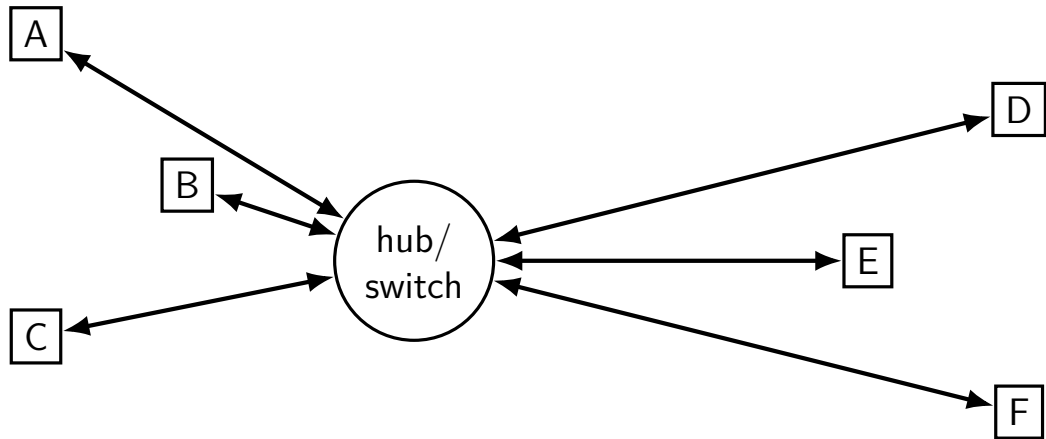
coordinating machine transmits timeslots

- part of common cellphone design (TDMA: time division multiple access)

make bus support multiple transmitters?

- requires understanding how interference works
- another part of common cell phone design

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what does the hub do?

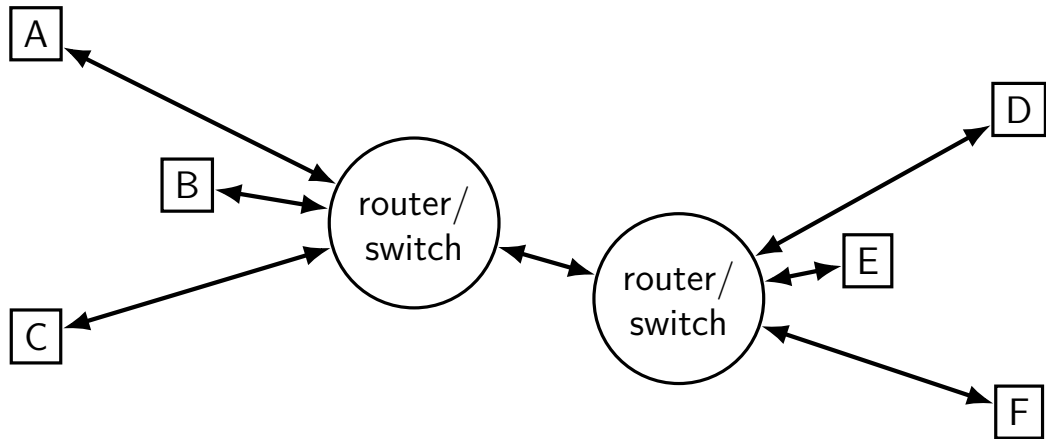
simple version:

- imitate shared bus: copy messages to everyone else
- something to handle two messages sent at once

less simple:

- read “header” on message + send to destination only
- requires some way to figure out destinations
- queue of messages waiting to be sent

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more complicated designs

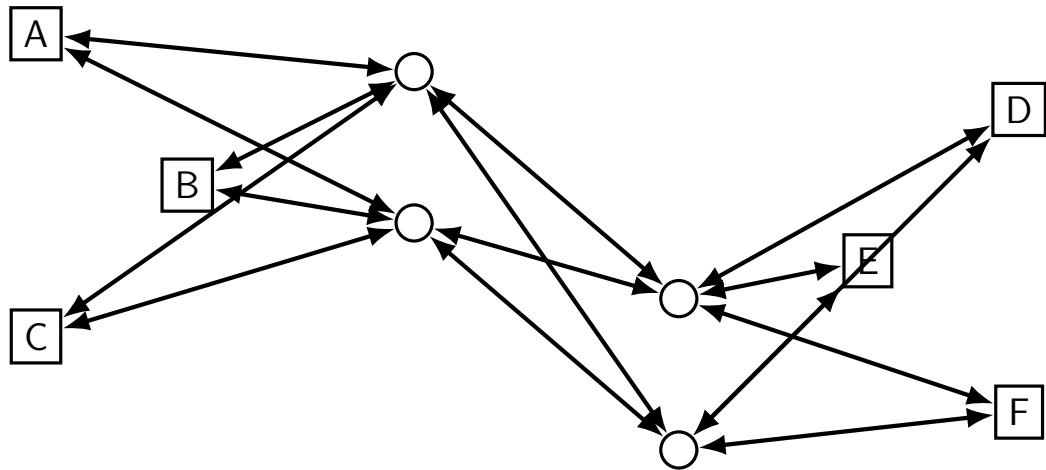
hierarchies

networks of networks

“internetworks”

so far still have single points of failure

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individual computers are networks

individual computers are (kinda) networks of...

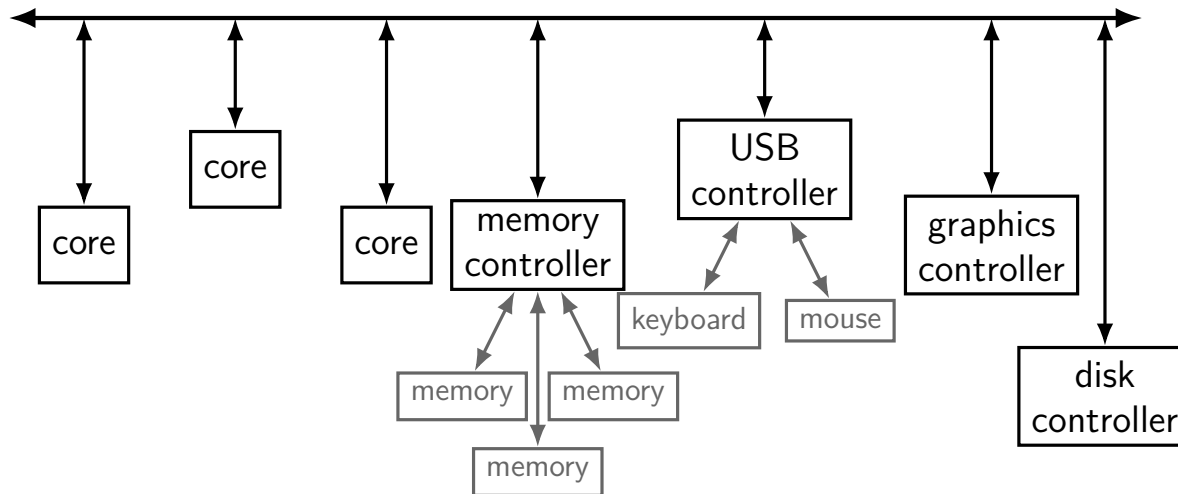
- processors

- memories

- I/O devices

so what topology (layout) do those networks have?

the “bus”



example: 80386 signal pins

name	purpose	
CLK2	clock for bus	timing
W/R#	write or read?	metadata
D/C#	data or control?	
M/IO#	memory or I/O?	
INTR	interrupt request	
...	other metadata signals	
BE0#-BE3#	(4) byte enable	address
A2-A31	(30) address bits	
DO-D31	(32) data signals	data

example: AMD EPYC (1 socket)

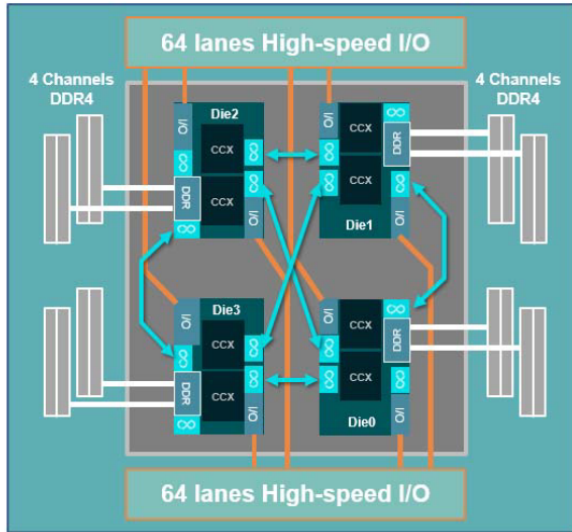


Fig. 21. Single-socket AMD EPYC™ system (SP3).

Figure from Burd et al,
" 'Zeppelin': An SoC for Multichip Architectures" (IEEE JSSC Vol 54, No 1)

example: Intel Skylake-SP

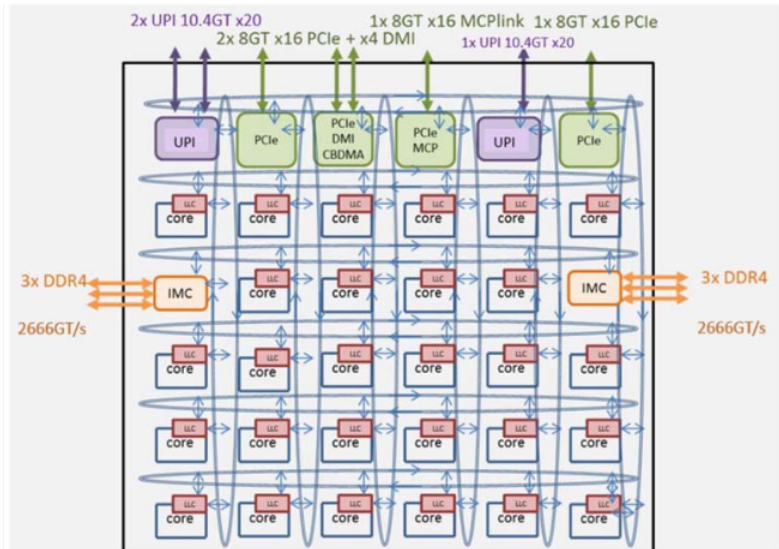
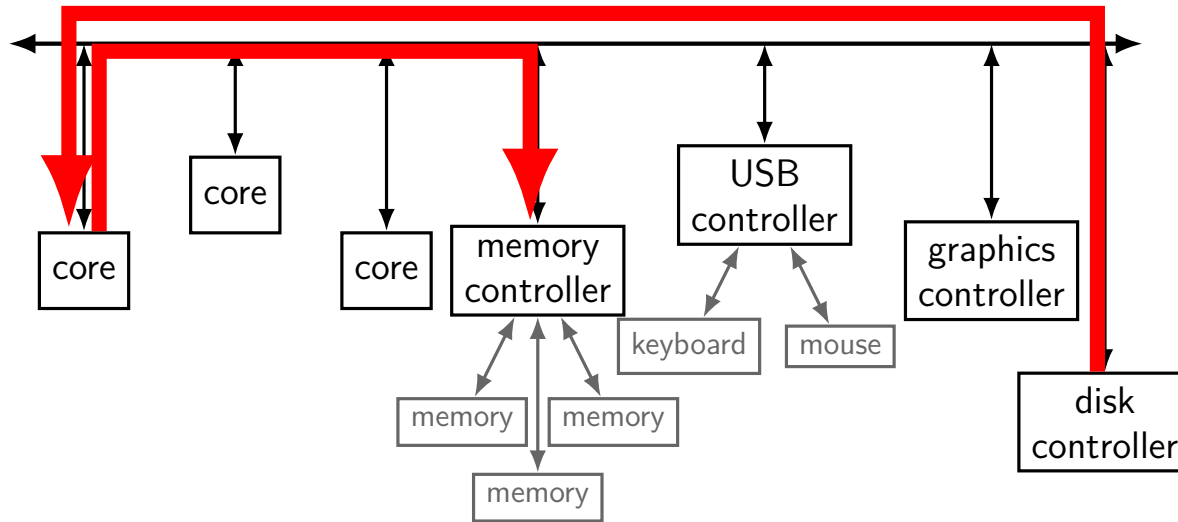
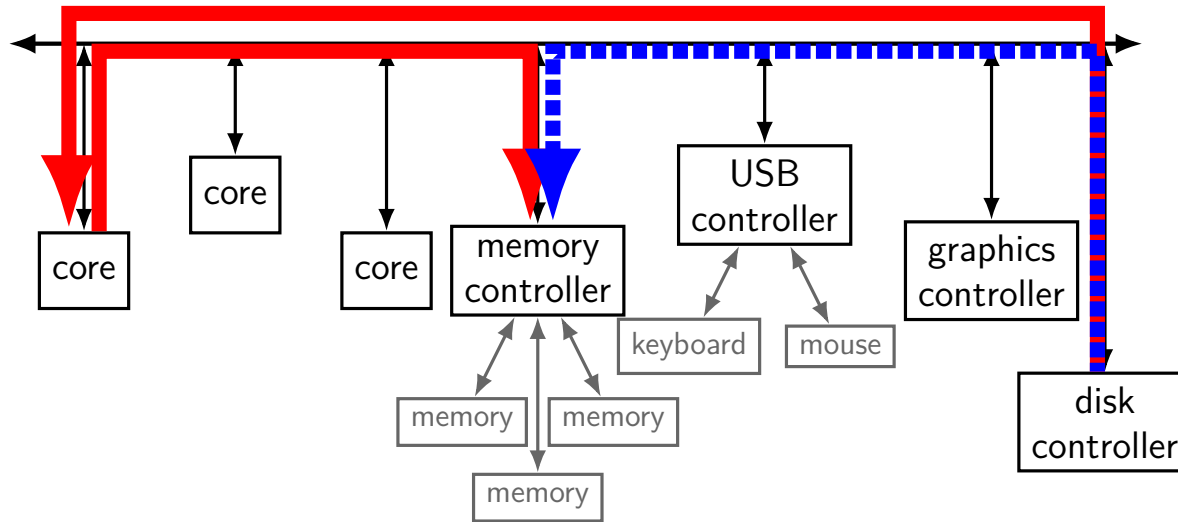


Figure from Tam et al, "SkyLake-SP: A 14nm 28-Core Xeon® Processor" (ISSCC 2018)

extra trips to CPU

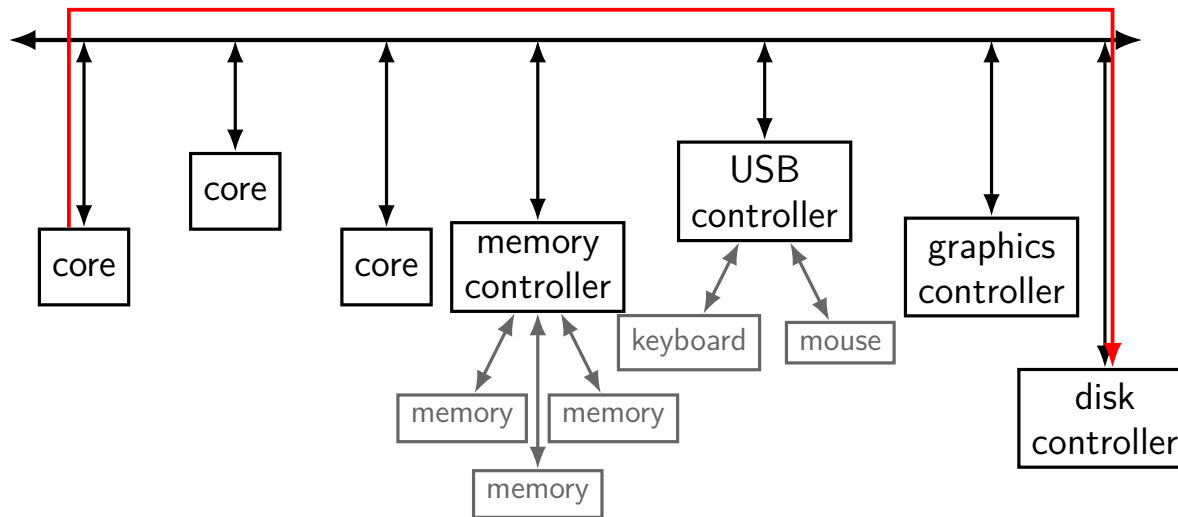


extra trips to CPU



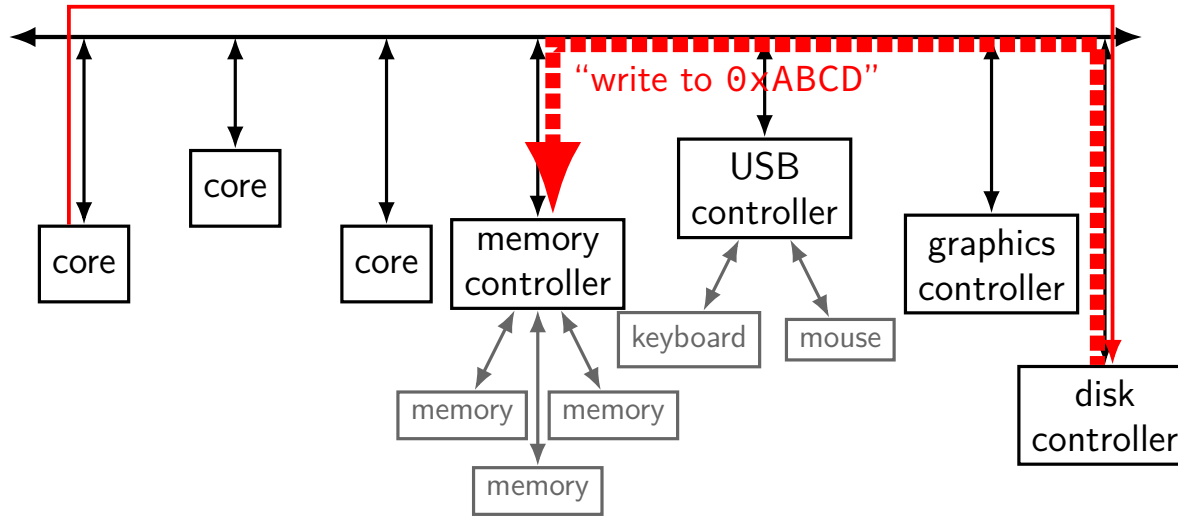
DMA

“place data at 0xABCD”



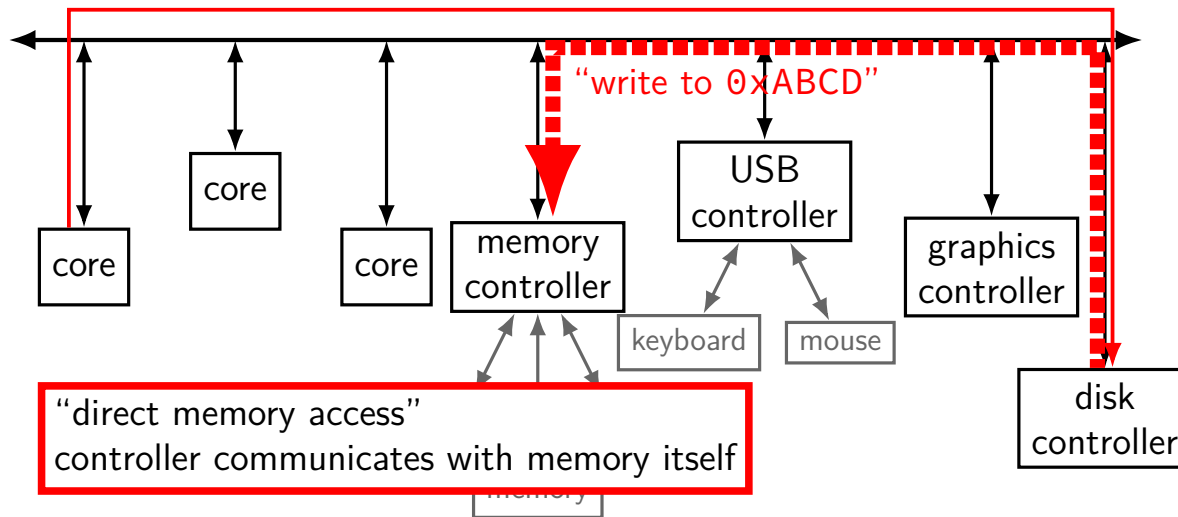
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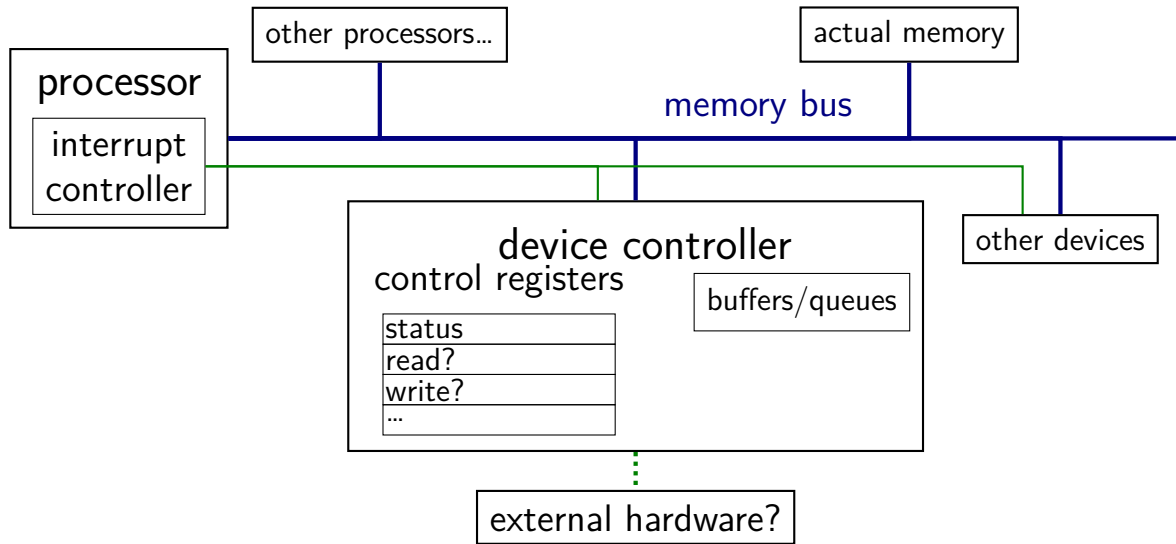
DMA

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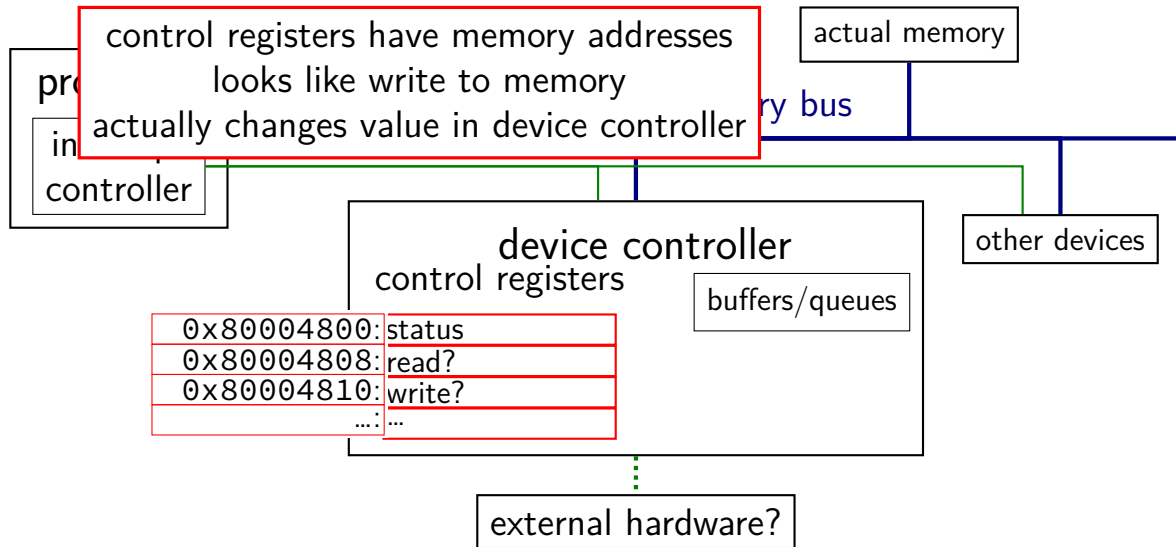


backup slides

connecting devices

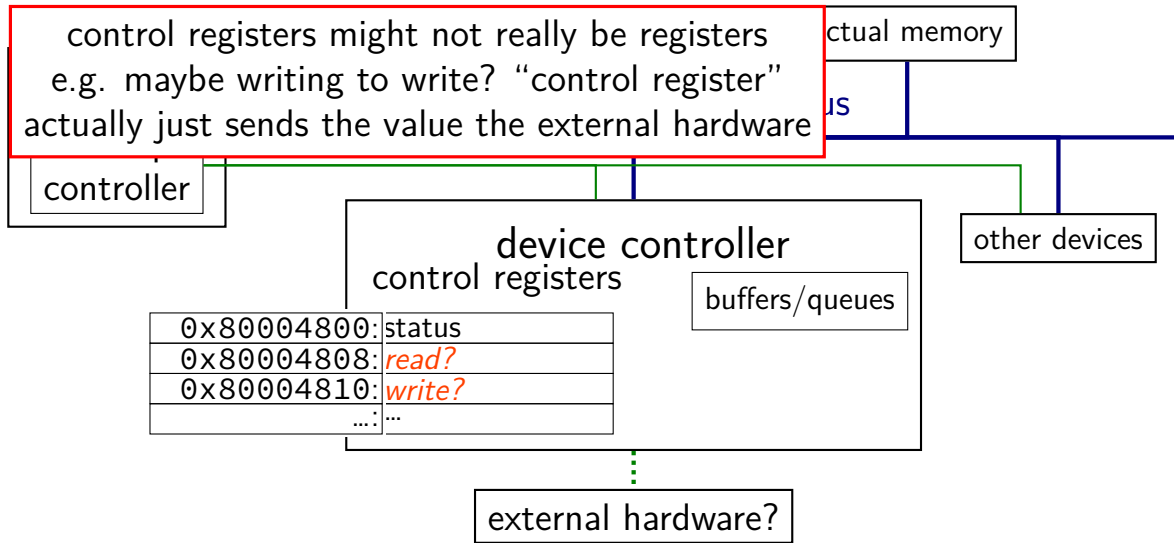


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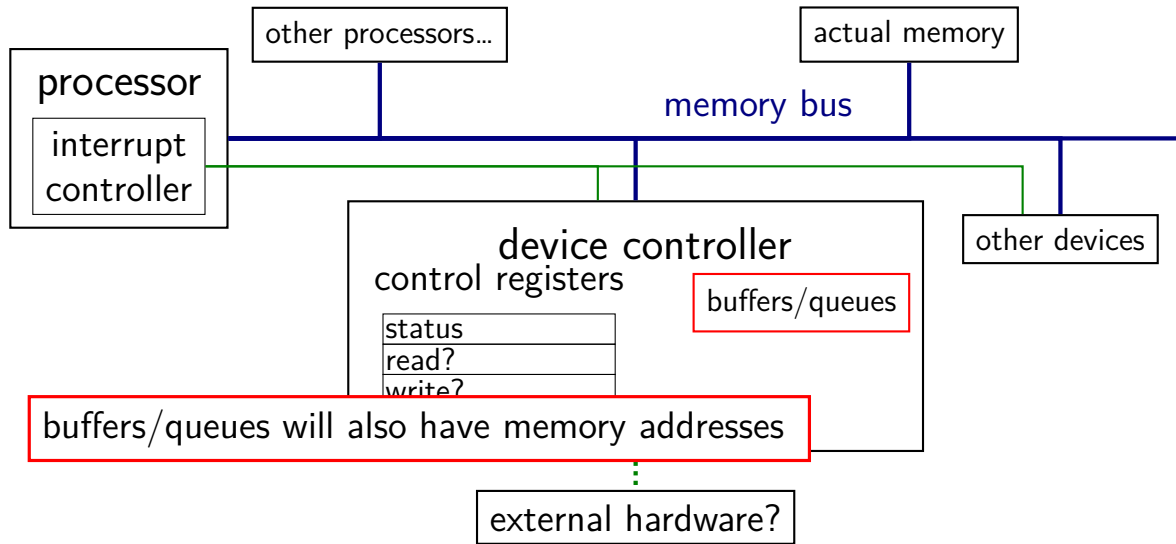


connecting devices

control registers might not really be registers
e.g. maybe writing to write? “control register”
actually just sends the value the external hardware



connecting devices



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